DATA SCIENCE PRACTITIONER - LEARNER WORKBOOK

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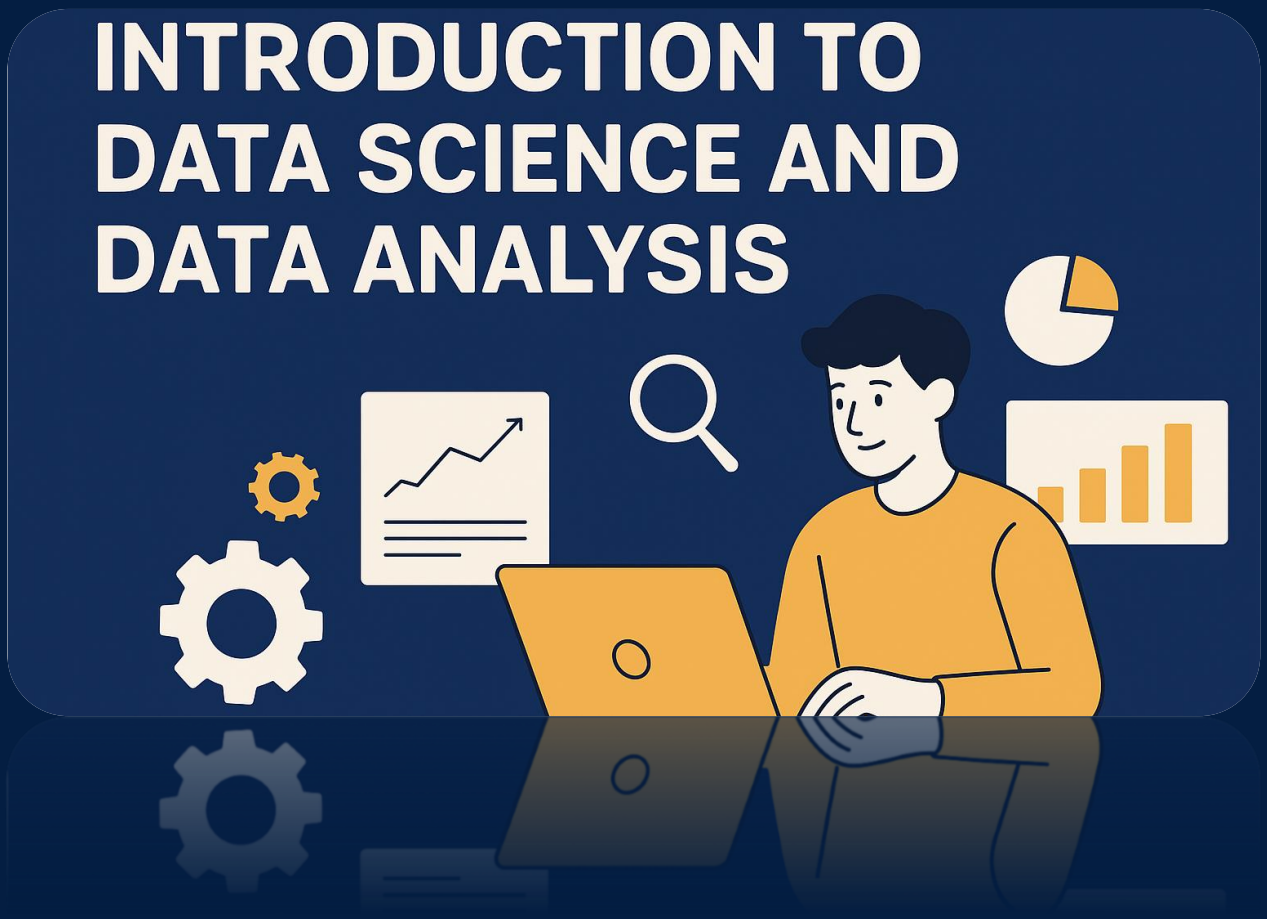
Connecting data, people, and the cloud.

KNOWLEDGE MODULE 01: INTRODUCTION TO DATA SCIENCE AND DATA ANALYSIS



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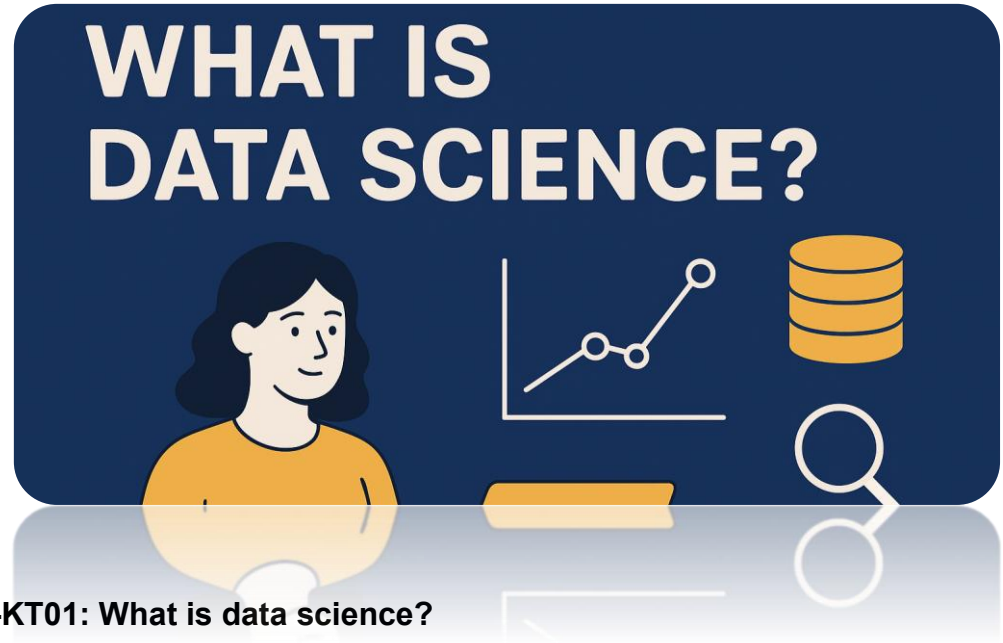

KM-01-KT01: What is data science?

1. **What is business intelligence? 3 marks**

Business Intelligence (BI) is the process of performing descriptive analysis on data using technology and skills to make informed business decisions. BI tools collect, govern, and transform data to enable the sharing of insights with stakeholders. It helps organizations understand market trends, identify revenue opportunities, improve processes, and stay competitive.

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2. **What Is Data Analysis? 5 marks**

Data analysis is the process of collecting, modelling, and analyzing data to extract meaningful insights that support decision-making. It involves cleaning, aggregating, and interpreting data using mathematical and statistical techniques to discover patterns and relationships. The goal is to convert raw data into actionable information for strategic and operational decisions.

3. **Describe current global trends in Big Data and explain their impact. 12 marks**

Big Data refers to large and complex datasets that traditional tools cannot process efficiently.

Cloud computing enables scalable storage and real-time analytics, reducing costs.

AI and Machine learning integration allow predictive and prescriptive analysis in real time.

Internet of things growth adds massive streams of sensor data used for automation and smart systems.

Data ethics and privacy laws (GDPR, POPIA) improve trust and responsible use of data.

Industry adoption: Compamies such as Netflix and Amazon use Big Data to personalize services and optimize operations.

Global impact: Organizations derive competitive advantages, governments make data-driven policies, and consumers receive smarter products and services.

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KM-01-KT02: Why data science?

1. **Explain how data science helps the economy. 6 marks**

Data science helps the economy by improving decision-making, productivity, and innovation across sectors such as finance, government, and industry. It enables economists to apply theoretical models to real-world data, making policies more evidence-based.
Through large-scale data analysis, economies can:

- Forecast trends and economic growth.
- Identify areas for investment and resource allocation.
- Improve public services and infrastructure through data-driven insights. Overall, it drives efficiency, competitiveness, and sustainable development.

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2. **Describe how data science helps society and communities. 6 marks**

Data science improves society by enabling data-driven solutions for social and humanitarian challenges.

It helps non-profits and governments design better programmes for healthcare, education, housing, and poverty reduction.

- Predictive models can save lives, for example, by identifying disease outbreaks or optimising resource distribution.
- It improves access to social benefits, speeds up refugee resettlement, and enhances community safety.
  In short, data science empowers organisations to make decisions that improve people's quality of life.

3. **Describe how data science helps businesses. 4 x 2 = 8 marks**

Data science supports businesses by enabling data-driven. Efficient, and predictive operations.

1. Increases business predictability:
Uses machine learning and analytics to forecast demand, sales, and customer behaviour.
Example: Pick n Pay uses weather and sales data to stock correctly before peak periods.

2. Ensures real-time intelligence:
Dashboards and live data feeds allow managers to act instantly.
Example: Uber Eats adjusts routes dynamically based on traffic and order volume.

3. Improves marketing and sales:
Analyses customer behaviour to deliver personalised promotions.
Example: Takealot launches flash sales based on trending products.

4. Enhances data security**:**
Detects fraud and system vulnerabilities early.
Example: Standard Bank uses algorithms to flag suspicious transactions.

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KM-01-KT03: Sources of data

1. **Explain the concept of data capture. 3 marks**

Data capture is the process of collecting information from various sources and converting it into a digital format that computers can process and analyze. it ensures that data from forms, sensors, or other inputs is gathered accurately for use in decision-making, reporting, or automation.

2. **Define a data capture system. 4 marks**

A data capture system is a set of tools or technologies used to collect, record, and store data automatically or manually from different sources. it converts real-world information such as typed forms, scanned barcodes, voice commands, or sensor readings into usable digital data. without these systems, organizations could not effectively track, analyze, or manage their operations.

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3. **List and describe different methods of capturing data. 3 x 3 = 9 marks**

Method 1: manual entry
data is entered by a person using a keyboard, form, or spreadsheet. examples include recording customer information or sales transactions manually.

Method 2: Optical character recognition (OCR)
Software extracts printed or handwritten text from scanned documents or images. For example, banks use OCR to capture details from ID documents for verification.

Method 3: barcode or QR code scanning
A scanner reads encoded information on products or tickets. this is common in retail stores, warehouses, and logistics operations to speed up checkout and inventory control.

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KM-01-KT04: Ensuring access to accurate data

1. **What is electronic identity verification? 6 marks**

Electronic identity verification, or EIDV, is the process of confirming a person's identity by matching their personal details—such as name, ID number, and address—against trusted digital databases. it uses online systems and biometric tools like facial recognition to verify that someone is who they claim to be without needing to visit an office. EIDV ensures security, prevents fraud, and allows businesses to comply with laws like POPIA and FICA.

2. **Why is identity verification software Important? 3 x 2 = 6 marks**

Security and fraud prevention:
Identity verification software helps detect fake identities and prevents fraudulent transactions, keeping users and organizations safe.

Improved customer experience:
It allows customers to verify their identities online quickly and easily without physical paperwork, speeding up onboarding and service delivery.

Regulatory compliance:
Industries such as banking and telecommunications must follow FICA, and KYC regulations. Verification software helps them meet these legal requirements accurately and efficiently.

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3. **Explain document data capture with an example. 3 marks**

Document data capture is the process of using technology to extract key information from digital or scanned documents automatically. for example, when a customer uploads a photo of their South African ID for mobile registration, the system uses optical character recognition (OCR) to read details such as full name, ID number, and date of birth, then stores them securely for verification.

4. **Identify the type of data that should be indexed. 3 x 2 marks**

Unique identifiers:
Used for quick lookups and linking records. examples include ID numbers, employee numbers, and passport numbers.

Dates and timestamps:
Used for filtering and tracking events over time, such as transaction dates or login times.

Categorical fields:
Used for grouping or filtering data by category, such as province, department, or account type.

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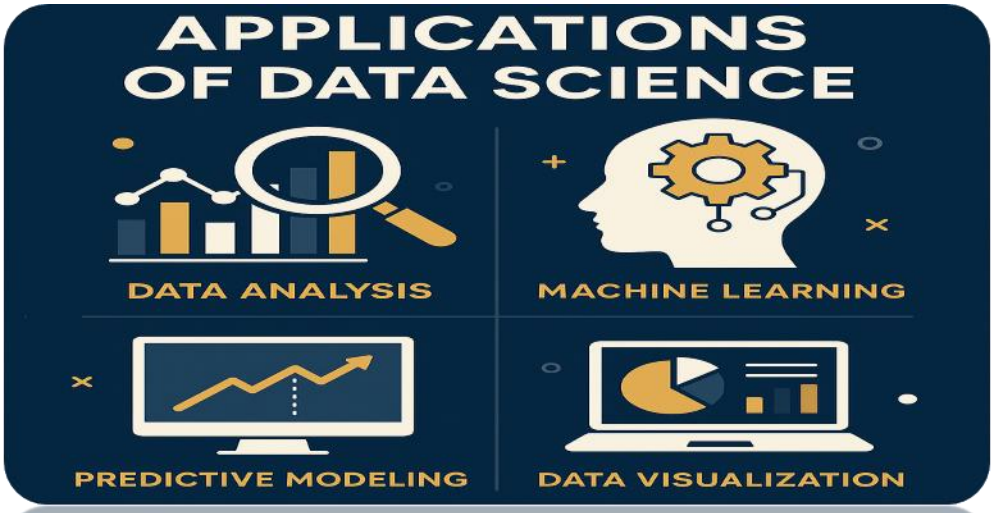

KM-01-KT05: Applications of data science

1. **Describe how data science supports decision-making in business. At least 3 examples x 2 = 6 marks**

Improving customer understanding:
Companies analyze customer behaviour to predict needs and preferences. for instance, Amazon and Zara use purchase and browsing data to recommend products that customers are most likely to buy. this increases sales and customer satisfaction.

Enhancing fraud detection:
Banks and financial services use machine learning to detect suspicious activity. PayPal and Visa analyze transaction patterns in real time to identify and block fraudulent actions, protecting customers and reducing losses.

Optimising operations and logistics:
Businesses use predictive analytics to improve efficiency. Uber and delivery companies use real-time data to optimize driver routes, reduce delivery time, and cut fuel costs.

Guiding strategic decisions:
Data dashboards and reports help executives make informed decisions based on evidence instead of guesswork. By analyzing trends and performance metrics, organizations can plan future actions confidently.

2.

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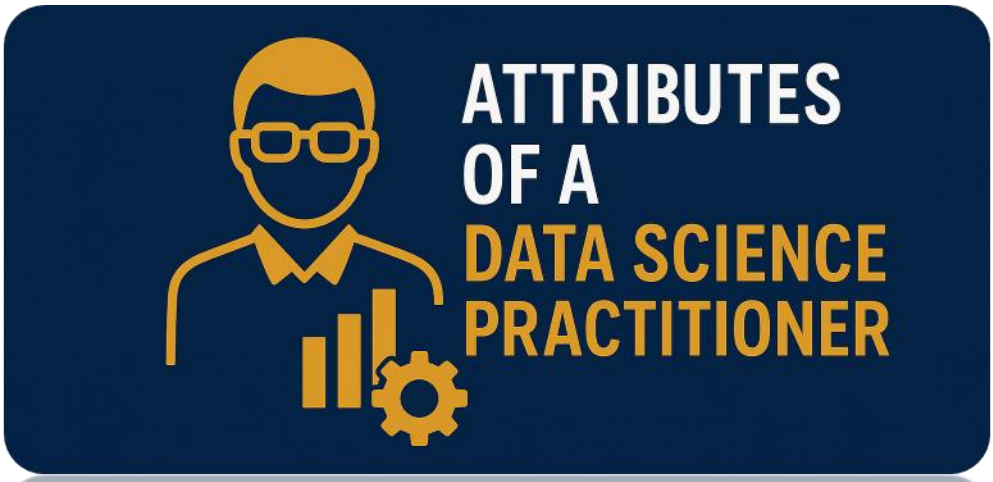

KM-01-KT06: Attributes of a Data Science Practitioner

1. **Explain how the following skills elevate the quality and impact of data science. 3 x 1 = 3 marks**

- Oral and written communication skills
- Attention to detail
- Analytical thinking

Oral and written communication skills:
Good communication allows data scientists to explain technical results in simple language that business teams can understand. This ensures insights are used effectively to guide decisions.

Attention to detail:
Data science requires accuracy when cleaning data, coding, and interpreting results. attention to detail helps avoid mistakes and ensures reliable outcomes.

Analytical thinking:
Analytical thinking helps data scientists break down complex problems, identify patterns, and design logical solutions using data. It improves the quality of insights and supports innovation.

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2. **Identify and explain the key attributes and attitudes required for a Data Science practitioner to progress in their career path. 3 x 2 = 6 marks**

Curiosity and continuous learning:
Data science is an evolving field. Successful practitioners stay curious and continually learn new tools, programming languages, and analytical techniques to remain relevant.

Problem-solving mindset:
A good data scientist approaches challenges with creativity and persistence. they use data to find practical solutions to real-world business or social problems.

Ethical responsibility and professionalism:
Data scientists must handle sensitive information responsibly, respecting privacy and fairness. maintaining ethical standards builds trust and strengthens credibility.

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KM-01-KT07: Big Data

1. **Describe the four vs of Big Data, i.e., volume, velocity, variety, and veracity. 4 x 3 = 12 marks**

Volume:
Refers to the massive amount of data generated every second from sources such as social media, sensors, and online transactions. Organizations store and process terabytes or petabytes of information to gain insights.

Velocity:
Refers to the speed at which data is created, collected, and processed. Examples include streaming data from IOT devices or live transaction systems that require real-time analysis.

Variety:
Represents the different types and formats of data available, including structured data (like spreadsheets), semi-structured data (like json), and unstructured data (like videos, images, and text).

Veracity:
Refers to the reliability and accuracy of the data. poor-quality or biased data can lead to incorrect conclusions, so it is important to verify data sources and maintain data integrity.

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2. **List and describe common sources of Big Data. 3 x 3 = 9 marks**

Social media platforms:
Sites like Facebook, Twitter, and Instagram produce vast amounts of data from user posts, likes, comments, and interactions. this data helps businesses analyze consumer sentiment and behaviour.

Internet of Things (IOT) devices:
Sensors and connected devices in homes, vehicles, and factories continuously collect data about temperature, location, usage, and performance, enabling automation and smart systems.

Transactional and business systems:
Systems such as point-of-sale terminals, online shopping carts, and banking databases record millions of transactions daily, generating valuable information about sales, spending, and customer preferences.

Public and government data:
Open data portals, census reports, and environmental sensors provide information for research, urban planning, and policy-making.

REFERENCE

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